

Introduction to Data Science

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Contents

- ◎ What is Data Science?
- ◎ Skills for Data Science
- ◎ Data Science Process
- ◎ Applications
- ◎ Why to learn?
- ◎ About course

What is Data Science?

◎ **Extracting useful conclusion** from large datasets with:

- Processes
- Problem definition
- Set of principles
- Analysis
- Algorithms
- Computer



Data Scientist vs all

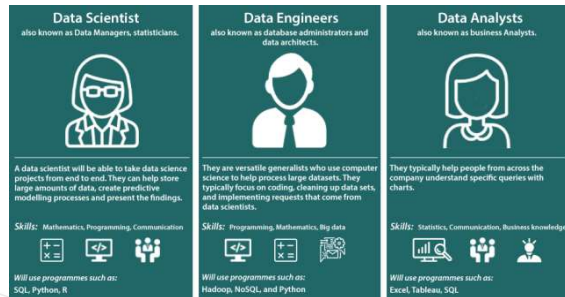


Data Scientist?

Data Scientist vs all

There's a joke that says:

"A data scientist is someone who knows more statistics than a computer scientist and more computer science than a statistician."



History of Data Science

History of data collection

- 3200 BC....
- Record experiences, events through writing
- Example: the sale of an item, the issuing of an invoice, the delivery of goods, ...

History of data analysis

- Statistics
 - Summary (descriptive) statistics
 - Probability theory, probability distributions
 - Statistical learning

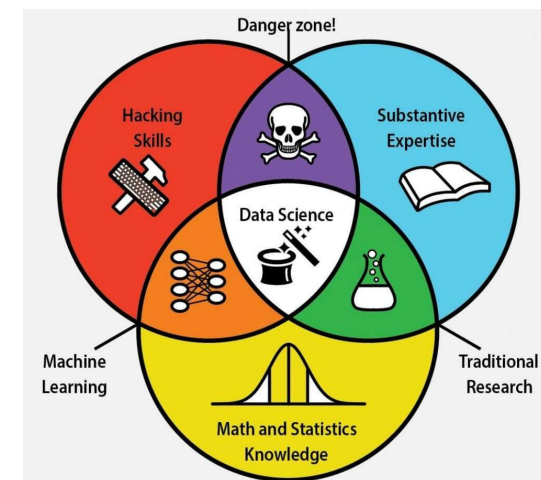
1990s: emergence and evolution of data science

The need for statisticians to join with computer scientists to bring mathematical rigor to the computational analysis of large data sets.

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Skills-set for a data scientist



Skills-set for a data scientist

Roles/Skills	Advanced Analytics	Business Acumen	Data Visualization	Communication and Collaboration	Software Skills
Data Analyst					
Business Analyst					
Data Scientist					
Data Architect					
Data Engineer					
Data Visualization Engineer					

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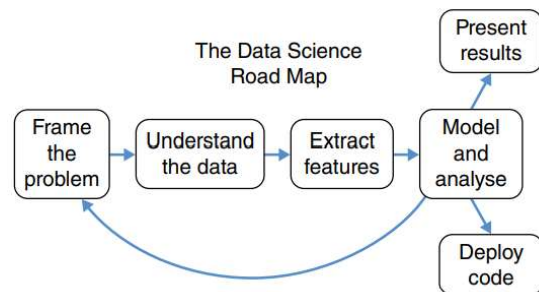
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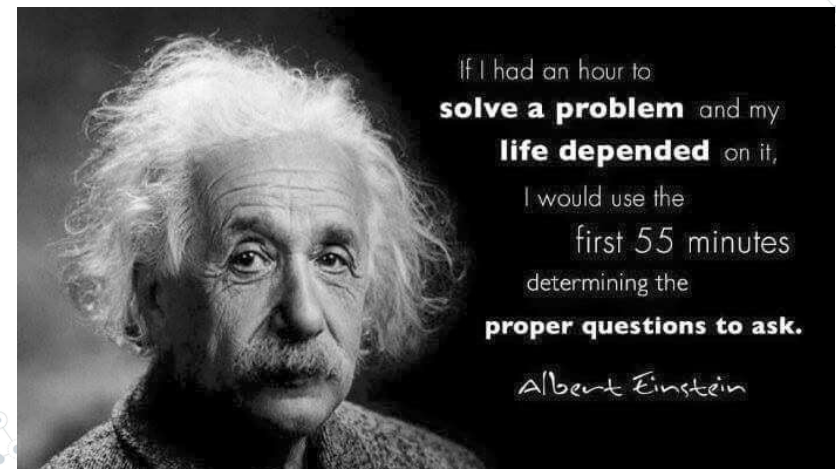
What is DS-ist do?



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Data Scientist ~ “Frame is problem”



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Example 1

- Let's go shopping!

Milk, eggs, sugar,
bread



Customer1

Milk, eggs, cereal,
bread

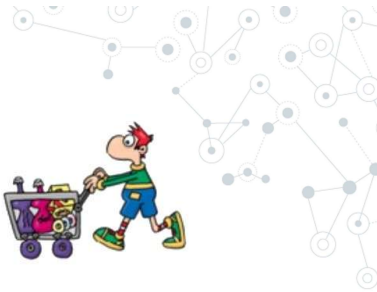


Customer2

Eggs, sugar



Customer3



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Example 2

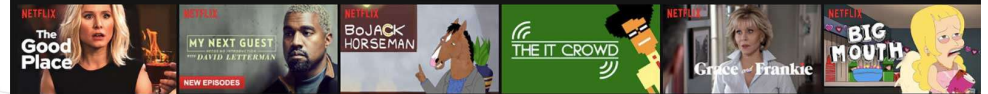
Emmy-winning US TV Shows



Police Detective TV Dramas

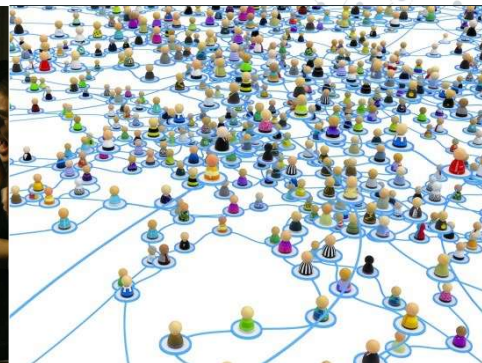


Critically Acclaimed Witty TV Shows



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Example 3



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Where Is Data Science Used?

Case studies that illustrate the impact of data science:

- Consumer companies: sales and marketing
 - Walmart
 - NetFlix
 - Amazon



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Where Is Data Science Used?

Case studies that illustrate the impact of data science:

- Governments
 - Organize cities: track, analyze, and control environmental, energy, and transport systems and to inform long-term urban planning
 - Design drugs for individual patients
 - Predict crime hot spots and recidivism

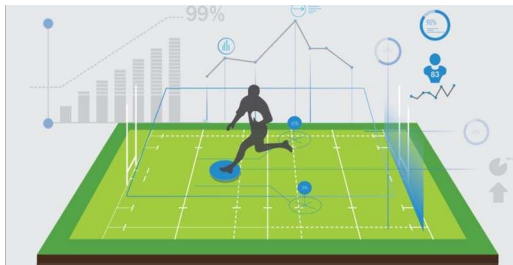


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Where Is Data Science Used?

Case studies that illustrate the impact of data science:

- Consumer companies: sales and marketing
- Governments: improve health, criminal justice, and urban planning
- Professional sporting:
 - Strategies



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Where Is Data Science Used?

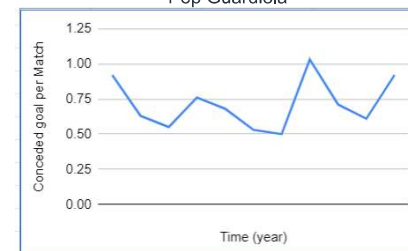
Case studies that illustrate the impact of data science:

- Professional sporting:
 - Strategies
 - Player recruitment

Team	Year	Conceded goal	Matches	Conceded/match
Barcelona (Spain)	2008	35	38	0.92
	2009	24	38	0.63
	2010	21	38	0.55
	2011	29	38	0.76
Bayern Munich (Germany)	2013	23	34	0.68
	2014	18	34	0.53
	2015	17	34	0.5
Manchester City (England)	2016	39	38	1.03
	2017	27	38	0.71
	2018	23	38	0.61
	2019	35	38	0.92



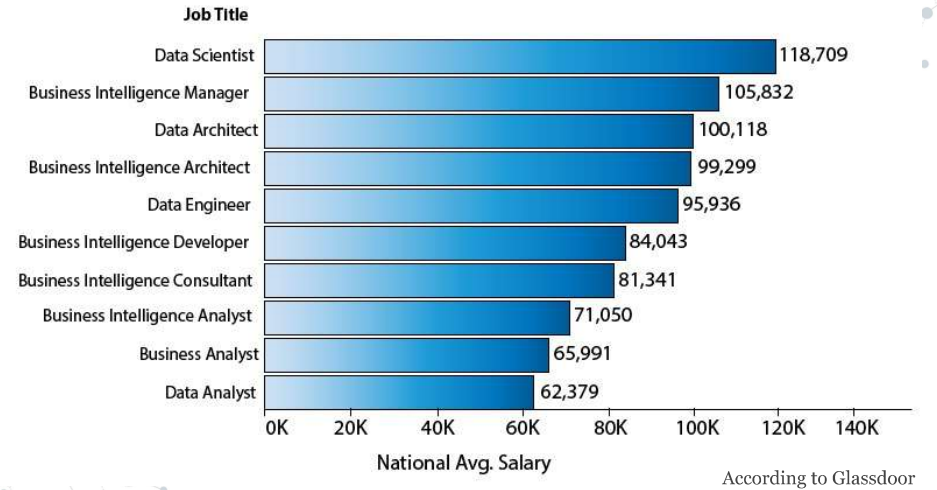
Pep Guardiola



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Why Learn Data Science?



Why Learn Data Science?

29 việc làm

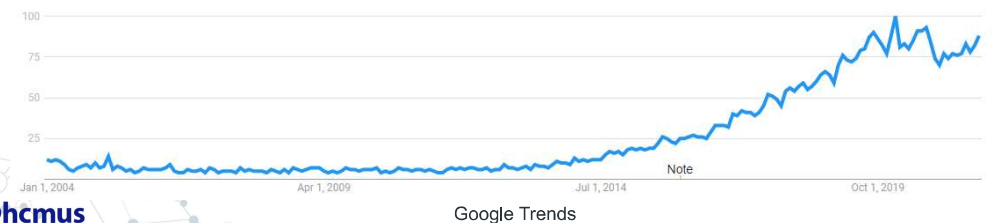
Data Scientist
Công Ty Cổ Phần Giáo Dục L...
\$ Lương: Dưới 55 Tr VND
Hồ Chí Minh
Xem video
Lưu việc làm
14/09/2021

Data Scientist
GalaxyOne
\$ Lương: Trên 22 Tr VND
Hồ Chí Minh
Lưu việc làm
07/09/2021

Senior PHP Developer
Công Ty TNHH MTV Wacontre
\$ Lương: 28,6 Tr - 33 Tr VND
Hồ Chí Minh
Lưu việc làm
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Why Learn Data Science?

- Main reasons:
 - Gain problem solving skills
 - High demand
 - Analytics is everywhere, everything is data-driven
 - Lots of high paying job opportunities



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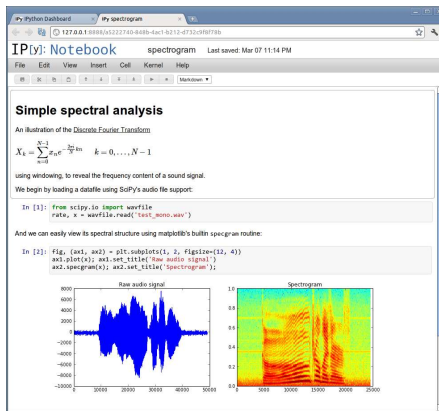
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Course Assessment

- **Contacts:**
 - Le Ngoc Thanh and tutors
 - Communication channel: Moodle, Email, Social Network
 - **Course Assessment:**
 - Final Examination: 50%
 - Exercises/Homeworks: 10%
 - Projects: 40%
 - **Regularizations:**
 - Attend classes regularly and have discussions
 - All material copied or amended from any source (e.g. internet, books) must be referenced correctly according to the reference style you are using.
- Any plagiarism will be 0 points for course grade.

Tools

- Python and the Libraries
- IPython Notebook



References

- [MIT \(Edx\), The Analytics Edge \(2018\)](#)
- [Harvard, Introduction to Data Science \(2018\)](#)
- [Berkeley, The Foundations of Data Science](#)
- [Berkeley, Principles and Techniques of Data Science](#)
- [Andrew Ng \(Coursera\), AI For Everyone](#)
- [Caltech, Learning from Data](#)